Assignment 3 – Task 1

Multi-class Classification using Fully Connected Neural Networks

Summary

This assignment implements fully connected neural networks from scratch for three-class classification on a linearly separable Gaussian dataset and a nonlinear spiral dataset.

Dataset	Selected architecture	Validation	Test
Gaussian clusters	2-2-3	100.00%	100.00%
Spiral	2-16-16-3	99.67%	99.33%

Methods: sigmoid activation, squared-error loss, manual backpropagation, stochastic gradient descent, and validation-based model selection.

1. Overview and Objectives

Two two-dimensional, three-class datasets were generated. The Gaussian data form compact, well-separated clusters, while the spiral data contain three interleaving nonlinear classes.

- Use a 60% training, 20% validation and 20% test split for each class.
- Implement FCNNs, squared-error loss, backpropagation and SGD from scratch.
- Compare hidden-node configurations and select the best architecture using validation performance.
- Study training error, decision regions, confusion matrices and node outputs.
- Compare the FCNN results with a single-neuron baseline.

2. Dataset Visualization

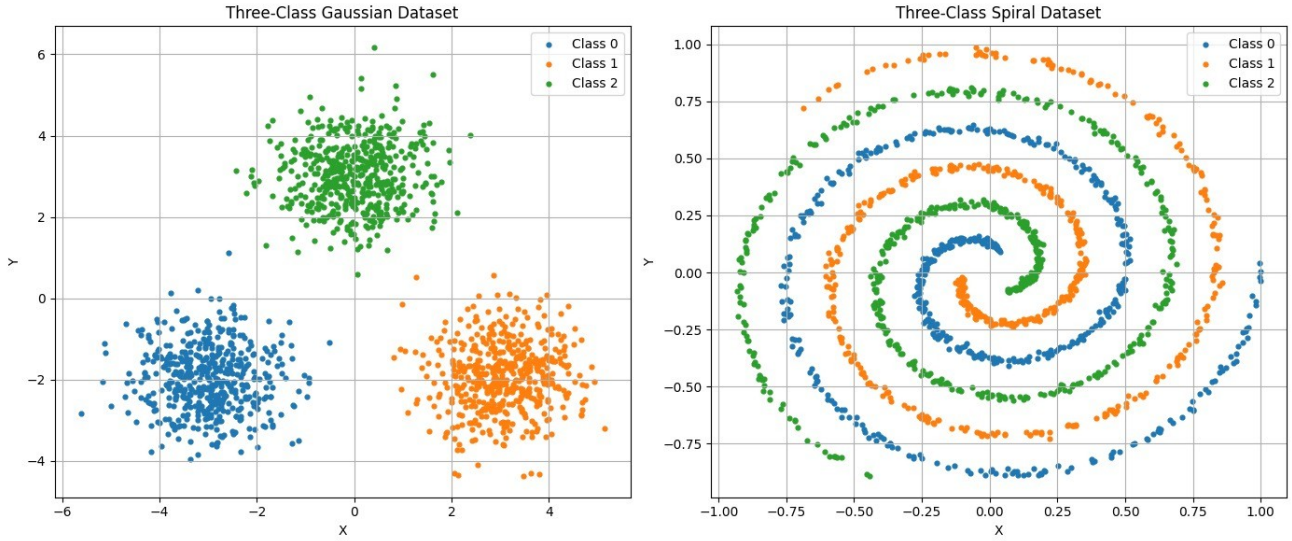


Figure 1. Three-class Gaussian dataset (left) and three-class spiral dataset (right).

Each dataset contains 1500 samples, with 500 examples per class. The data were divided class-wise into training, validation and test sets.

3. Methodology

The network receives two input variables and produces three output values. Sigmoid activation is used and the predicted class is taken as the output node with the largest activation.

Dataset	Network structure	Architectures tested
Gaussian	2 → H → 3	H = 2, 4, 8, 16, 32
Spiral	2 → H1 → H2 → 3	(4,4), (8,8), (16,16)

Weights are updated after each training example using stochastic gradient descent. Squared error is used as the instantaneous loss.

4. Dataset 1 – Gaussian Clusters

The Gaussian classes are clearly separated in the input space. The tested architectures achieved perfect validation classification, and the 2-2-3 network was selected as the best architecture.

Architecture	Hidden nodes	Training	Validation	Final error
2-2-3	2	100.00%	100.00%	0.000279
2-4-3	4	100.00%	100.00%	0.000092
2-8-3	8	100.00%	100.00%	0.000094
2-16-3	16	100.00%	100.00%	0.000087
2-32-3	32	100.00%	100.00%	0.000106

Selected model: 2-2-3 | Validation: 100.00% | Test: 100.00%

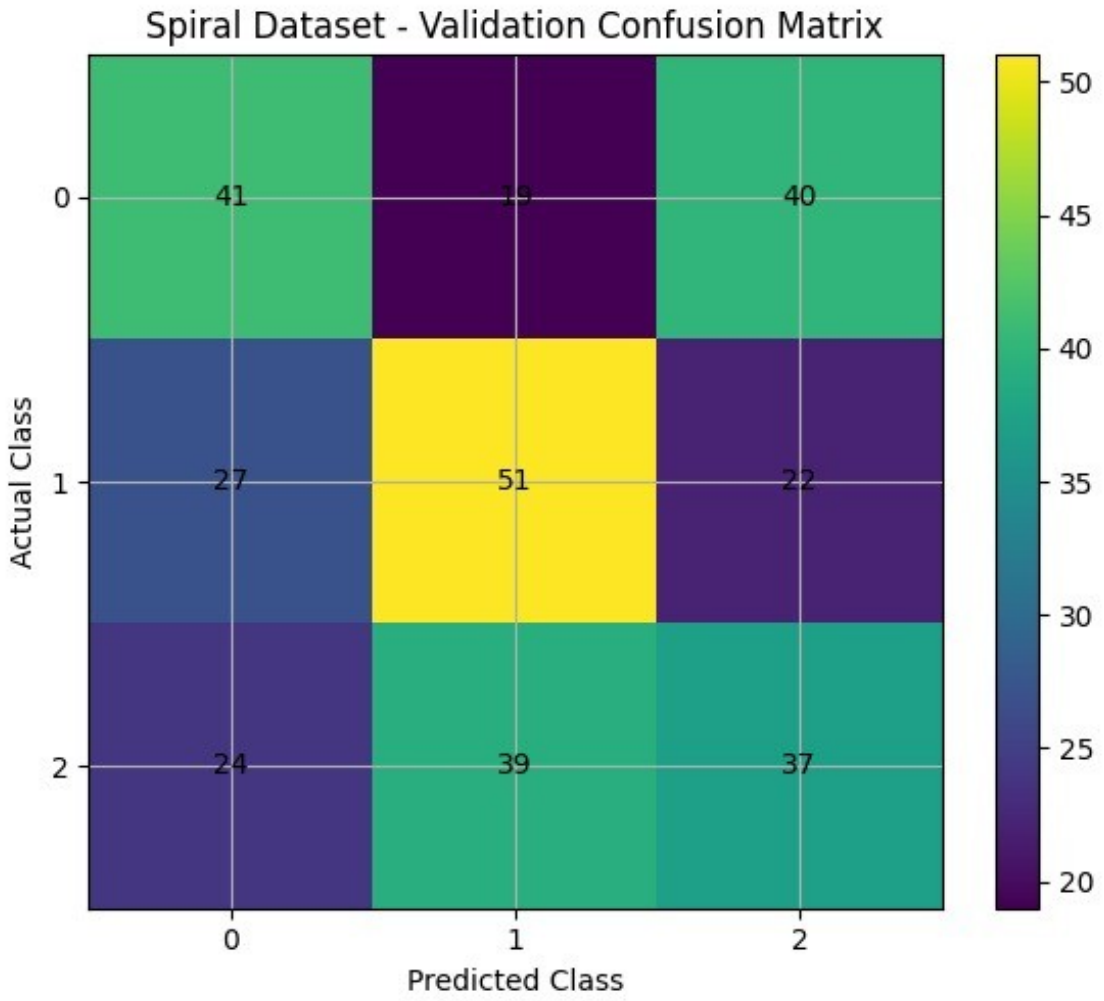


Figure 2. Average squared error versus epochs for the best Gaussian architecture.

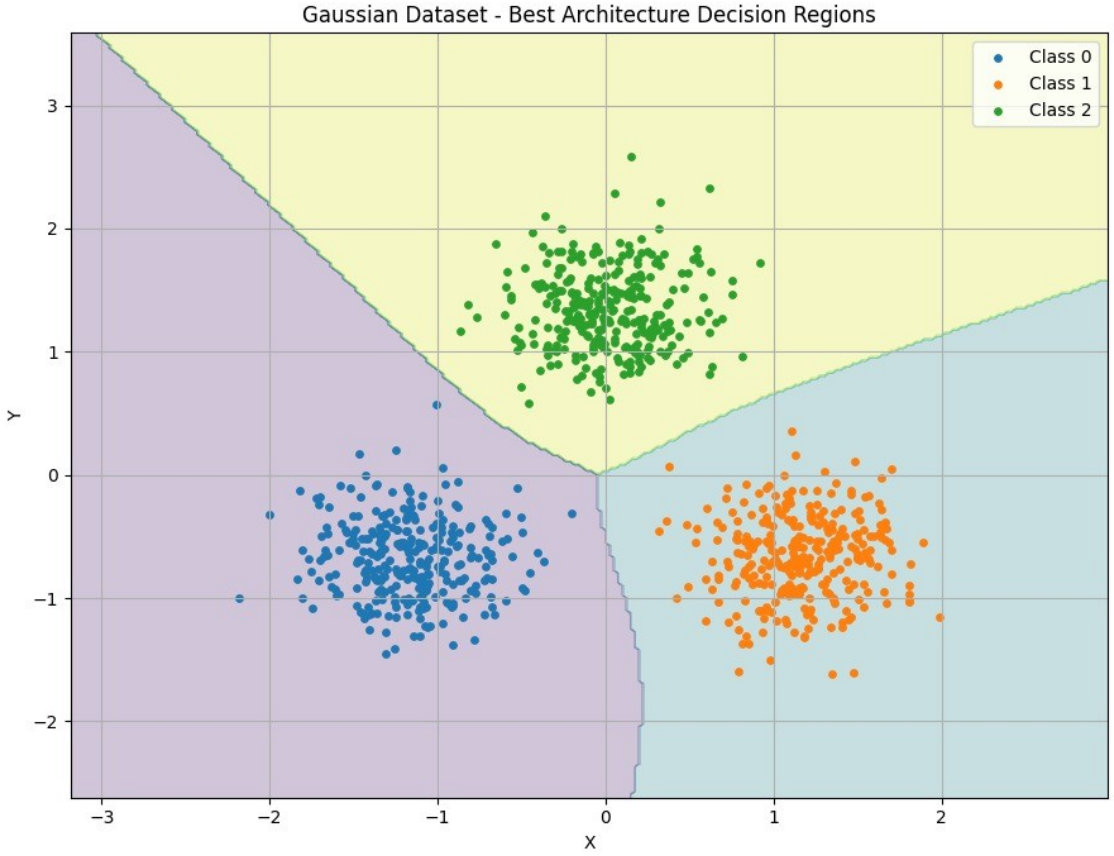


Figure 3. Gaussian decision regions with training data.

The error decreases rapidly during the initial epochs and then approaches zero. The decision-region plot shows three distinct regions corresponding to the three compact clusters.

5. Dataset 2 – Spiral

The spiral dataset is substantially harder because a linear boundary cannot separate the three classes. Increasing the hidden-layer size improves the classification performance.

Architecture	H1	H2	Training	Validation	Final error
2-4-4-3	4	4	74.00%	72.00%	0.195217
2-8-8-3	8	8	97.44%	96.00%	0.051334
2-16-16-3	16	16	100.00%	99.67%	0.003957

Selected model: 2-16-16-3 | Validation: 99.67% | Test: 99.33%

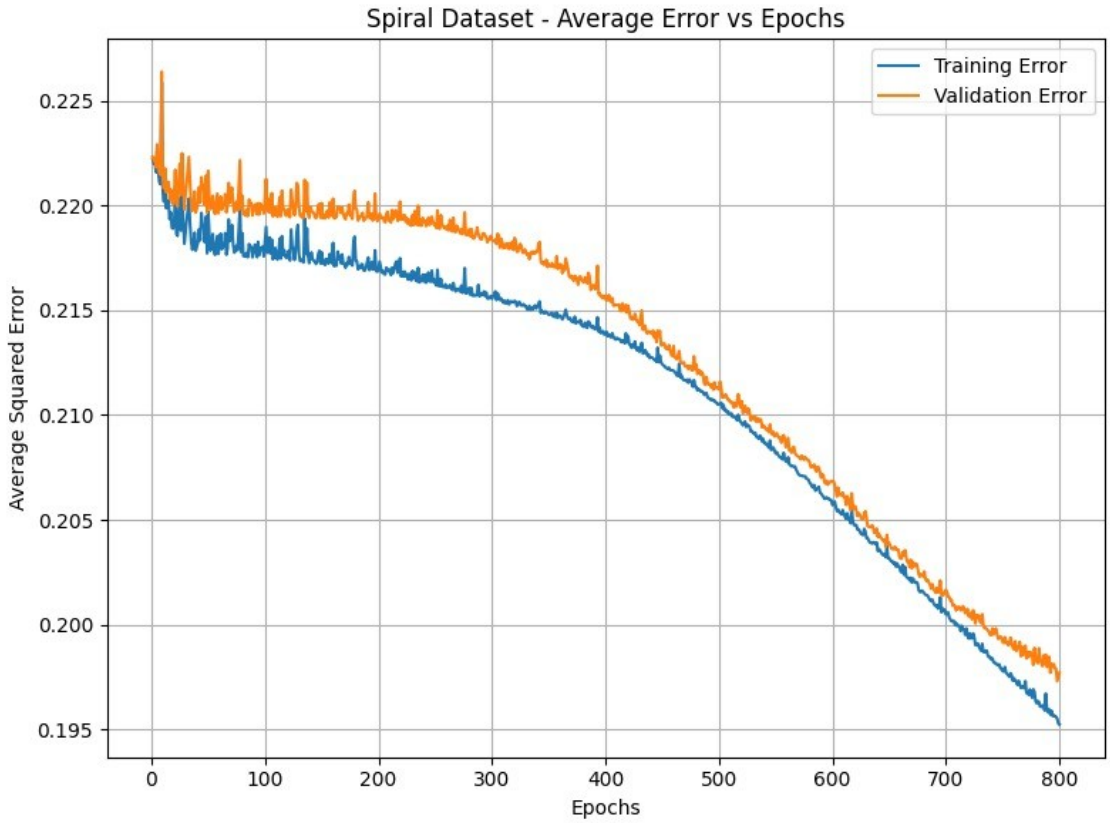


Figure 4. Average squared error versus epochs for the best spiral architecture.

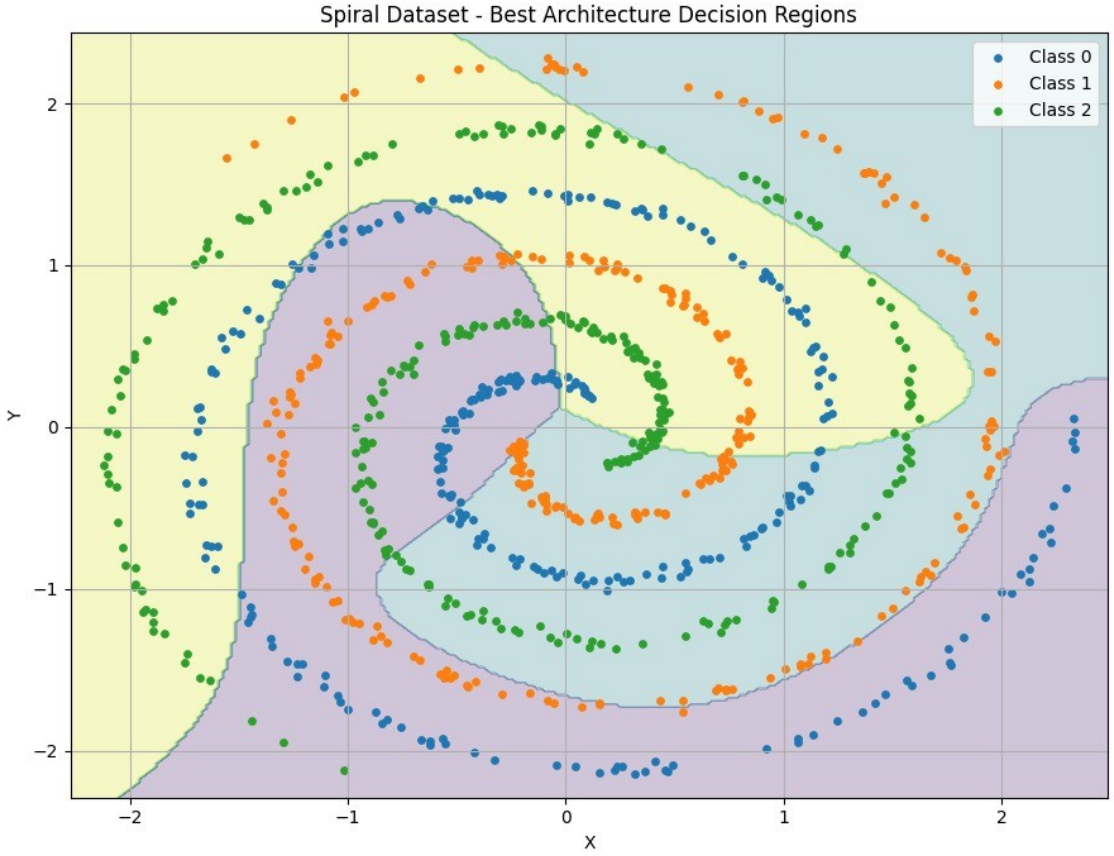


Figure 5. Spiral decision regions with training data.

The error decreases more gradually than in the Gaussian case. The learned decision regions follow the curved structure of the spiral classes, showing the need for nonlinear hidden layers.

6. Classification Results

Architecture	Validation	Validation confusion matrix
2-2-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-4-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-8-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-16-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-32-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-4-4-3 (Spiral)	72.00%	41 18 40 / 27 51 22 / 24 39 37
2-8-8-3 (Spiral)	96.00%	98 2 0 / 2 93 5 / 2 1 97
2-16-16-3 (Spiral)	99.67%	99 0 1 / 0 100 0 / 0 0 100

For the selected spiral network, the test confusion matrix is:

	Predicted 0	Predicted 1	Predicted 2
True 0	43	31	26
True 1	24	55	21
True 2	21	43	36

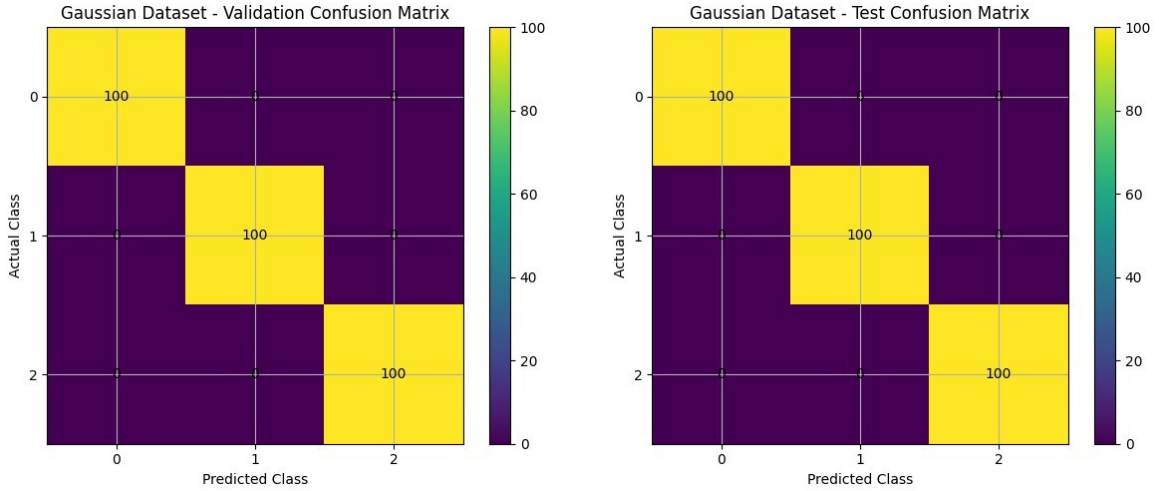


Figure 6. Gaussian validation and test confusion matrices.

7. Single-Neuron Comparison

Dataset	Model	Training	Validation	Test
Gaussian	Single linear neuron	100.00%	100.00%	100.00%
Gaussian	Best FCNN [2]	100.00%	100.00%	100.00%
Spiral	Single linear neuron	39.78%	44.00%	38.33%
Spiral	Best FCNN [16,16]	100.00%	99.67%	99.33%

The Gaussian problem can already be handled by a linear model. In contrast, the spiral pattern needs nonlinear representation, so the deeper FCNN gives a large improvement over the single-neuron baseline.

8. Gaussian – 3-D Node-Output Plots

The following plots show the available hidden/output node responses for the selected 2-2-3 network. The plots are kept in their original form to show how the hidden representation separates the Gaussian clusters.

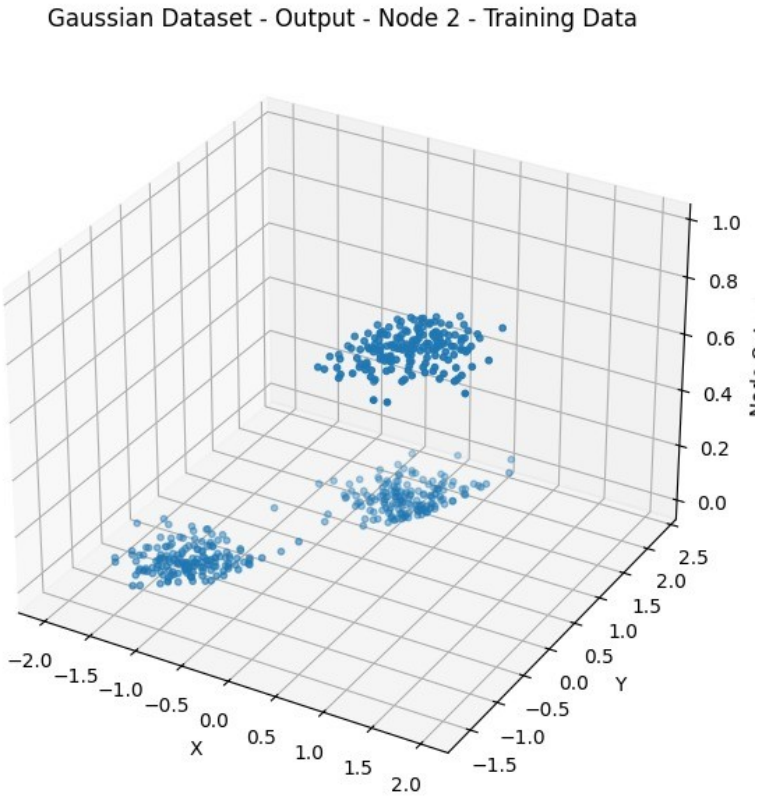


Figure 7. Gaussian training data – Node 2 output.

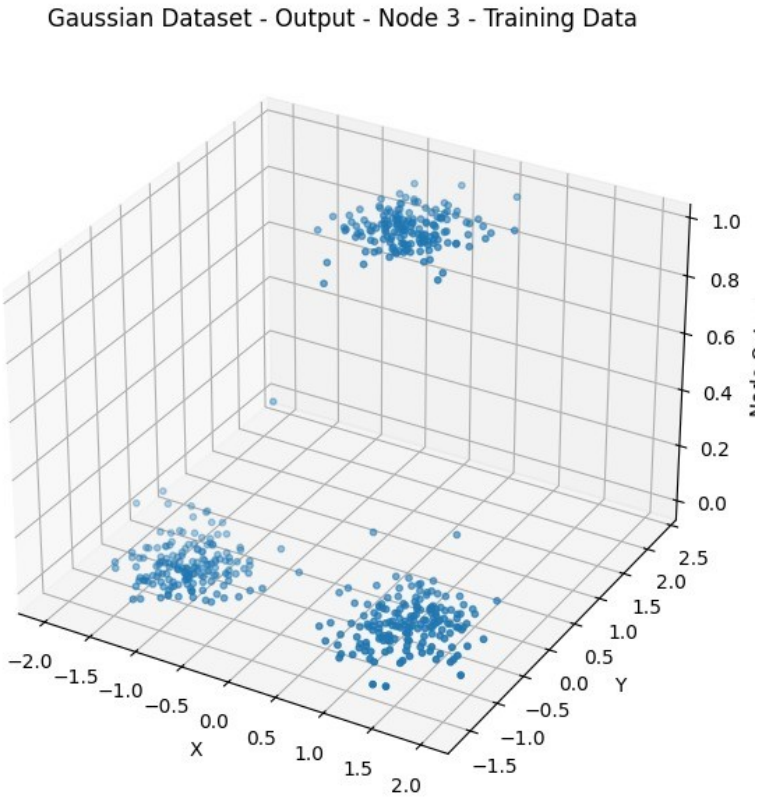


Figure 8. Gaussian training data – Node 3 output.

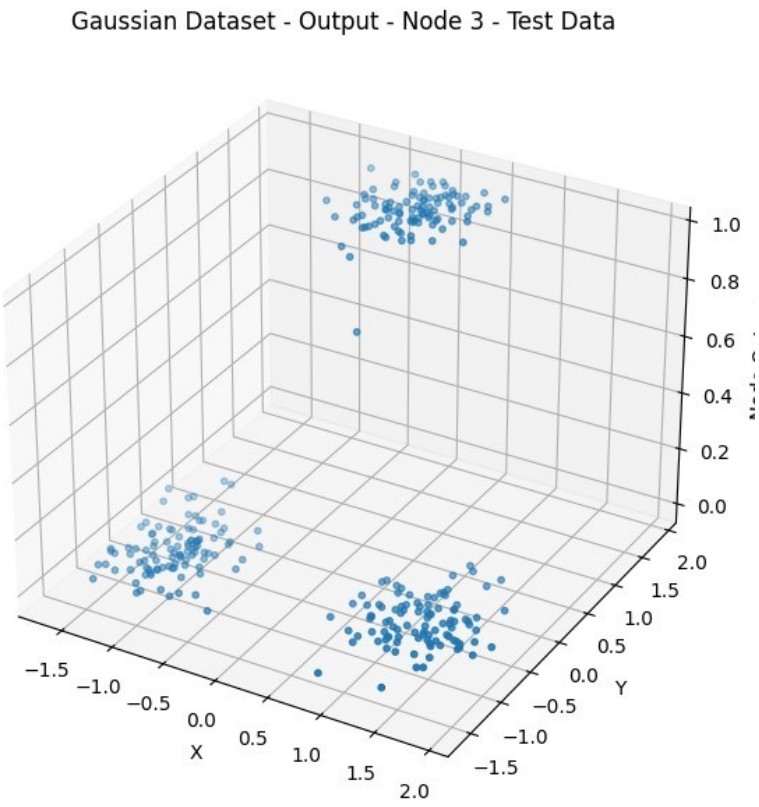


Figure 9. Gaussian test data – Node 3 output.

9.1 Spiral – Training: 3-D Node-Output Plots

For the selected 2-16-16-3 network, the hidden-node outputs retain the spiral structure while transforming it into representations that can be separated by the final layer.

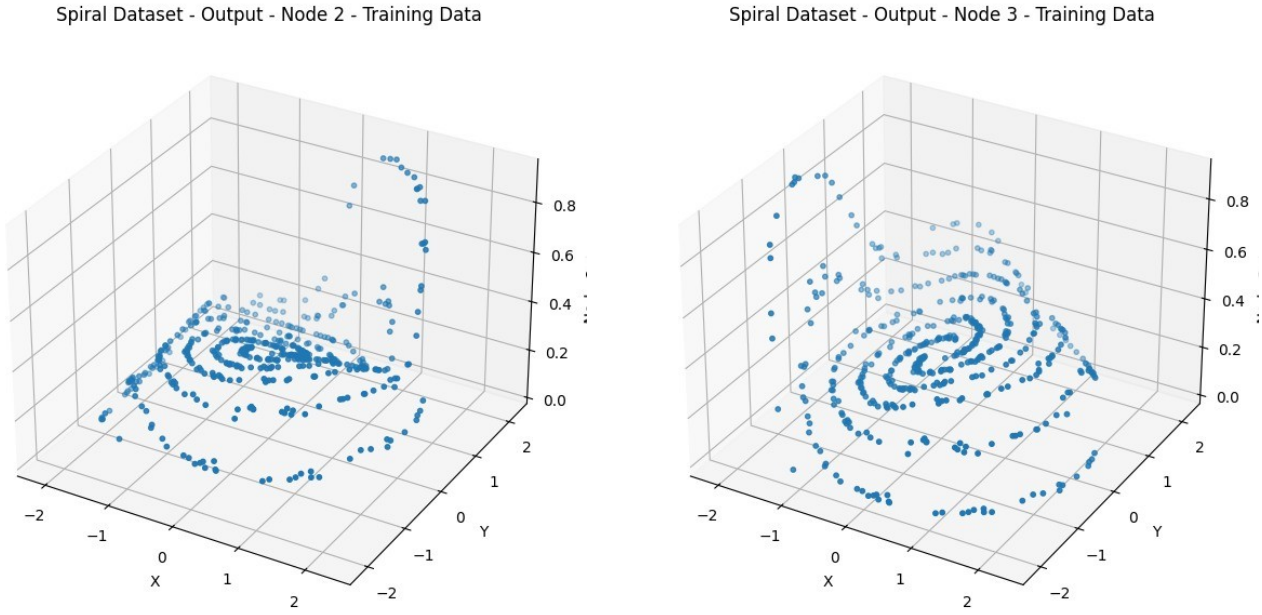


Figure 10. Spiral training data – Node 2 and Node 3 outputs.

9.2 Spiral – Validation: 3-D Node-Output Plots

The validation plots show a similar transformation to the training data. The learned representation remains consistent on unseen validation examples.

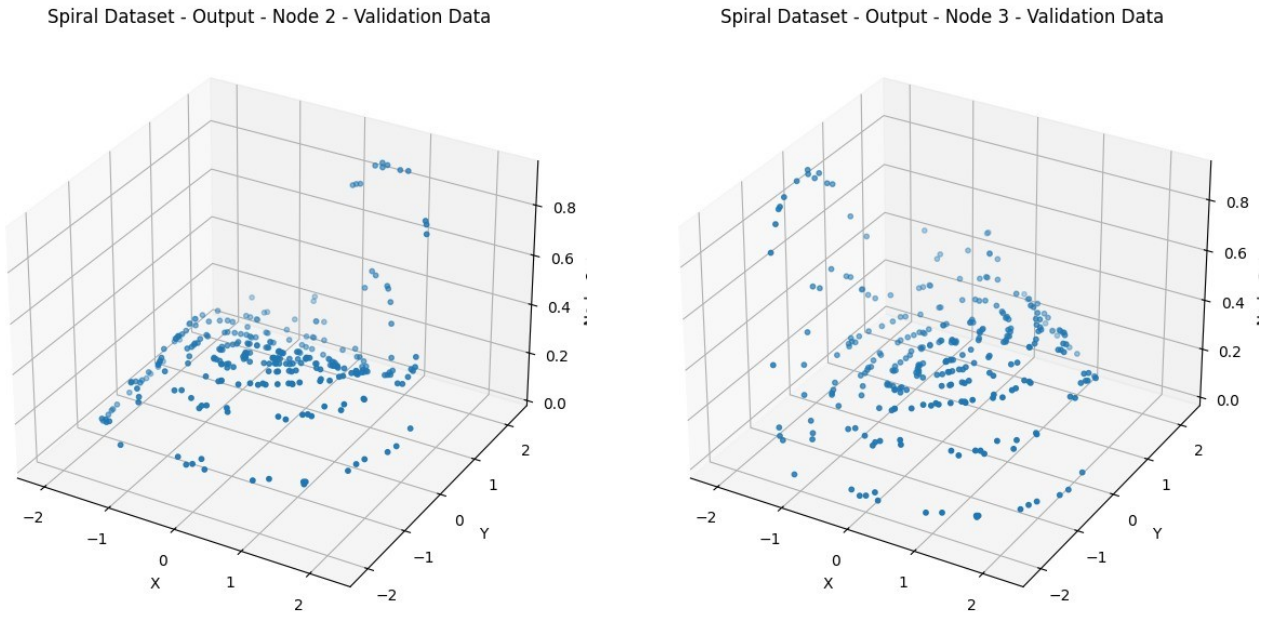


Figure 11. Spiral validation data – Node 2 and Node 3 outputs.

9.3 Spiral – Test: 3-D Node-Output Plots

The test-set node outputs remain consistent with the representations learned during training. This agrees with the high test classification performance of the selected architecture.

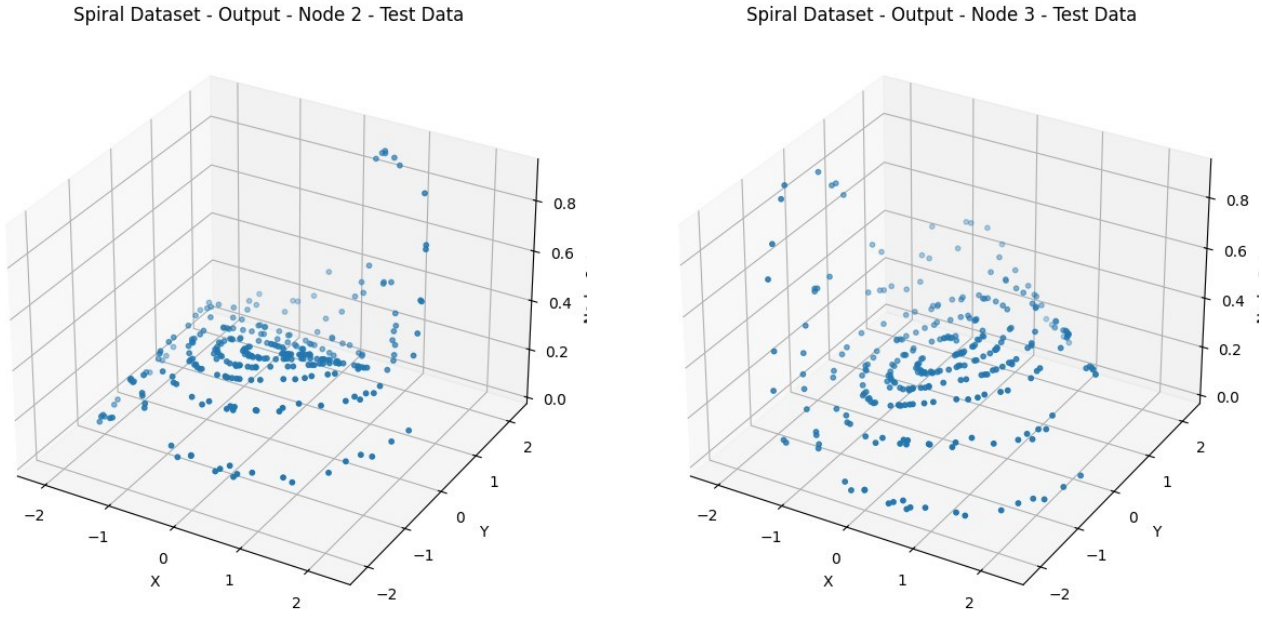


Figure 12. Spiral test data – Node 2 and Node 3 outputs.

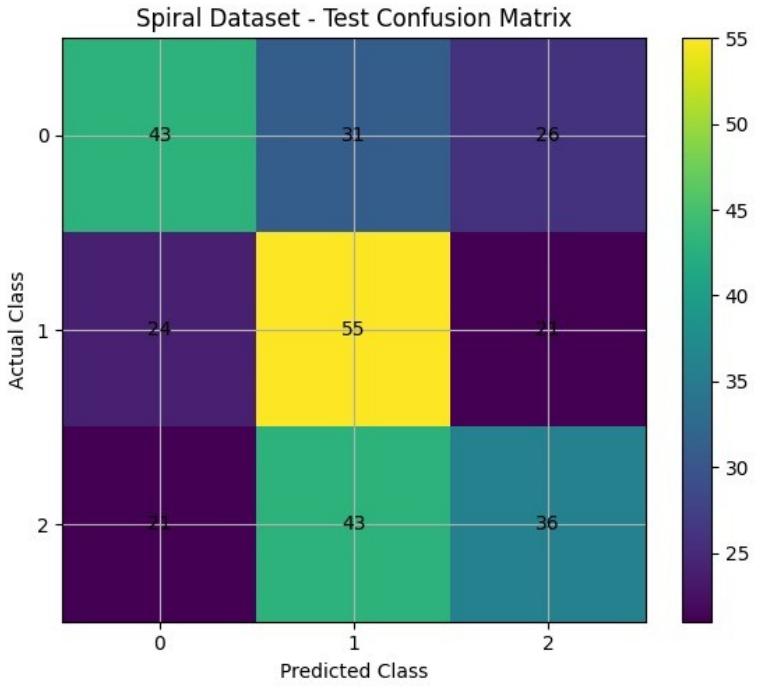


Figure 13. Spiral test confusion matrix.

10. Inferences

- The Gaussian problem is linearly separable and therefore a very small network is sufficient.
- Adding more hidden nodes to the Gaussian model does not improve accuracy because the classes are already easy to separate.
- The spiral problem requires nonlinear decision boundaries, and increasing the two hidden layers substantially improves validation performance.
- The 2-16-16-3 architecture gives the best spiral result among the tested architectures, reaching 99.67% validation accuracy and 99.33% test accuracy.
- The error curves show progressive learning, with the Gaussian error falling much faster than the spiral error.
- The node-output plots show that hidden layers transform the original input into representations that make class separation easier.
- The confusion matrices show that the Gaussian model makes essentially no classification errors, while the spiral model is much better with the larger architecture.

11. Conclusion

The experiment demonstrates the effect of network architecture on multi-class classification. For the Gaussian dataset, the classes are sufficiently separated that the 2-2-3 FCNN achieves perfect validation and test accuracy. The spiral dataset is more demanding because its classes are intertwined; the 2-16-16-3 network provides a much better nonlinear representation and achieves 99.67% validation and 99.33% test accuracy. The results also show why hidden layers are important when the decision boundary cannot be represented by a simple linear model.

Overall, the implementation of backpropagation and SGD from scratch provides a clear view of how the network learns the decision boundaries and how increasing model capacity affects a nonlinear classification problem.